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| Team Number:    | apmcm253XXXXX |
| Problem Chosen: | A             |

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### 2025 APMCM summary sheet

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Basic  $\text{\LaTeX}$  knowledge is assumed (references, equations, figures, tables, etc.). All commonly used packages are pre-loaded.

The latest template is maintained at <https://github.com/latexstudio/APMCMThesis>.

**Keywords:**  $\text{\LaTeX}$  Mathematical Modeling APMCM Thesis Template

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## **I. Introduction**

To clarify the origin of the problem, the following background information is provided.

## **II. Problem Restatement**

### **2.1 How do we approximate the entire process?**

- Point 1
- Point 2
- Point 3

### **2.2 How do we define the optimal configuration?**

1. From the perspective of efficiency:
2. From the perspective of cost:
3. Trade-off between the two:

### **2.3 Local Optimization vs. Global Optimization**

- Local optimization
- Global optimization
- Practical balance

### **2.4 Differences in Weights and Sizes**

### **2.5 What if no data are available?**

## **III. Models**

### **3.1 Basic Model**

#### **3.1.1 *Notation and Definitions***

Most symbols and definitions are derived from queuing theory.

### **3.1.2 Assumptions**

1. Assumption 1
2. Assumption 2
3. Assumption 3

### **3.1.3 Model Formulation**

#### 1) Utility function

- Waiting cost:
- Congestion loss:
- Weight of each factor:
- Trade-off strategy:

#### 3) Global vs. local optimization

- Global optimization:
- Local optimization:
- Optimal number of servers:

### **3.1.4 Solution and Results**

1. Integer programming solution
2. Numerical results

### **3.1.5 Result Analysis**

- Comparison of local and global optimization
- Sensitivity analysis: results are sensitive to the three key parameters
- Parameter trends
- Comparison with existing designs

### **3.1.6 Strengths and Weaknesses**

**Strengths:** The improved model addresses factors neglected in the basic model. Results show it is more reasonable and significantly outperforms current designs.

**Weaknesses:** The model is still a large-scale approximation, limiting its use in high-precision scenarios.

## IV. Conclusions

### 4.1 Summary of Results

- Key finding 1
- Key finding 2
- Key finding 3

### 4.2 Methods Employed

- Queuing theory
- Integer programming
- Sensitivity analysis

### 4.3 Applications

- Application 1
- Application 2
- Application 3

## V. Future Work

### 5.1 Alternative Approaches

#### 5.1.1 *Limitations of Queuing Theory*

#### 5.1.2 *Other Potential Models*

#### 5.1.3 *Data-Driven Methods*

#### 5.1.4 *Dynamic Optimization*

## References

## VI. References

[1] Author(s), Book/Journal Title, Publisher/Journal, Year.

[2] L<sup>A</sup>T<sub>E</sub>X Resources and Tutorials, <https://www.latexstudio.net>

- [3] APMCMThesis Template Repository, <https://github.com/latexstudio/APMCMThesis>

## Appendix

Listing 1: MATLAB source code of the algorithm

```

kk=2; [mdd,ndd]=size(dd);
while ~isempty(V)
    [tmpd,j]=min(W(i,V)); tmpj=V(j);
    for k=2:ndd
        [tmp1,jj]=min(dd(1,k)+W(dd(2,k),V));
        tmp2=V(jj); tt(k-1,:)= [tmp1,tmp2,jj];
    end
    tmp=[tmpd,tmpj,j;tt]; [tmp3,tmp4]=min(tmp(:,1));
    if tmp3==tmpd
        ss(1:2,kk)=[i;tmp(tmp4,2)];
    else
        tmp5=find(ss(:,tmp4)~=0); tmp6=length(tmp5);
        if dd(2,tmp4)==ss(tmp6,tmp4)
            ss(1:tmp6+1,kk)=[ss(tmp5,tmp4);tmp(tmp4,2)];
        else
            ss(1:3,kk)=[i;dd(2,tmp4);tmp(tmp4,2)];
        end
    end
    end
    dd=[dd,[tmp3;tmp(tmp4,2)]]; V(tmp(tmp4,3))=[];
    [mdd,ndd]=size(dd); kk=kk+1;
end
S=ss; D=dd(1,:);

```